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Differentiation

• Inverse functions:

1) $\sin^{-1} n = y$

$\sin y = n$

$\cos y \frac{dy}{dn} = 1$

$$\frac{dy}{dn} = \frac{1}{\cos y}$$

$$= \frac{1}{\sqrt{1-n^2}}$$

2) $\cos^{-1} n = y$

$\cos y = n$

$-\sin y \frac{dy}{dn} = 1$

$$\frac{dy}{dn} = \frac{-1}{\sin y}$$

$$= \frac{-1}{\sqrt{1-n^2}}$$

• For hyperbolic Inverses, use the same method.

$\sinh^{-1} n = y$

$\therefore n = \sinh y$

$1 = \cosh y \cdot \frac{dy}{dn}$

$$\therefore \frac{dy}{dn} = \frac{1}{\cosh y} = \frac{1}{\sqrt{1+n^2}}$$

• Parametrics: $\frac{dy}{dn} = \frac{dy}{dt} \cdot \frac{dt}{dn}$, $\frac{d^2y}{dn^2} = \frac{d}{dt} \left(\frac{dy}{dn} \right) \cdot \frac{dn}{dt}$ ★

• Maclaurin's series:

$$y = f(0) + f'(0)n + \frac{f''(0)}{2!}n^2 + \frac{f'''(0)}{3!}n^3 \dots$$

Use calculator to confirm values of specific derivatives.